Course code: ME314

Course Name: Mechatronics

Pre-requisite: any subject on basic electronics, any subject on control, any subject on mathematical modeling, any subject on computer programming

Objective of the course:

In the increasingly competitive environment, mechatronics has become the key to industrial prosperity. The rapid advancements in the fields of electronics engineering, information technology and systems engineering have been responsible for evolving new concepts aimed at developing highly sophisticated machine tools for enhanced productivity. Mechatronics is the combination of electronics and computer technology to practical control applications in mechanical systems. Mechatronics is an exhaustive coverage of various aspects from design to testing high technology machines and systems. The objective of the course is to impart knowledge to the learners on mechanical systems, electrical systems, electronics systems, computer technologies, testing techniques and fault disgnosis techniques which are essential components of a mechatronic systems.

Course Content:

UNIT I: (3 Lectures)

Introduction to mechatronics, components, multidisciplinary nature, examples: camera, engine fuel control system

UNIT II: (12Lectures)

Basics of Electronics; semiconductor technology, transformer, transistor, integrated circuits (ICs), microcontrollers, circuits and programming

UNIT III: (6 Lectures)

Basic Mechanical systems; structural components, measurements, hydraulic and pneumatic systems, modeling techniques, controls

UNIT IV: (6 Lectures)

Electrical components; actuators, drives, dc motors, servo motors,

UNIT V: (5 Lectures)

Case study - CNC machine; components, functioning, programming

UNIT VI: (8 Lectures)

Artificial intelligence for mechatronics systems, neural networks, fuzzy logic, genetic algorithms

Laboratory Component: No

Readings:

Text Books:

1. Title: Mechatronics, Author: H. M. T. Ltd.
2. Title: Mechatronics principles and applications, Author: G. Onwubolu

Reference Books:

1. Title: Mechatronics, Authors: C. W. de Silva
2. Title: Mechatronics for beginners: 21 projects for PIC microcontrollers, Editors: A. Imam